Zoha Qamar

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Experience

Systems Developer — November 2025 - Present
Baffinland Iron Mines — *Oakville, ON*

- Develop distributed backend microservices using **Java Spring Boot**, **REST APIs**, and **Spring Security** for **Angular** applications, integrating **MS SQL Server**, stored procedures, and **SSMS** in scalable enterprise systems.
- Test, debug, deploy, and secure backend services using **Postman**, **PowerShell**, and remote server environments, resolving API integration, authentication, and production deployment issues.
- Collaborate in **GitLab** using **IntelliJ IDEA**, participating in code reviews and writing clean, maintainable, and well-documented production code within an Agile-style ticketing workflow.
- Developed internal automation tools and dashboard features that streamlined previously manual request-management workflows, improving data tracking, export processes, operational efficiency, and cross-team coordination.
- Led requirements gathering and established a systematic intake and planning process for future development requests, improving scope clarity, prioritization, and delivery efficiency across technical and non-technical stakeholders.

Application Programmer Co-op — May 2023 - August 2024
Ministry of Education — *Toronto, ON*

- Developed and tested web applications in an Agile team environment using **VBScript and HTML**, debugging and troubleshooting to improve system reliability and reduce post-deployment issues by **20%**.
- Automated testing workflows and accessibility compliance testing by integrating tools like WAVE, JAWS, and NVDA and ensuring **WCAG 2.0** adherence and enhancing usability for **1,000+ end users**.
- Authored **10+ detailed technical documents** covering workflows, testing protocols, and deployment steps, improving onboarding efficiency by **30%** and streamlining knowledge transfer across development teams.
- Tracked and resolved **40+ defects** using **ALM and JIRA**, collaborating across cross-functional teams to resolve **95%** pre-release and reduce defect resolution time, maintaining secure, high-quality codebases.

Education

University of Toronto — June 2025
Honours Bachelor of Science: Computer Science & Statistics

- **Relevant Courses:** Software Design, Theory of Computation, Data Structures & Analysis, Artificial Intelligence, Machine Learning, Databases, Probability & Statistics, Data Analysis, Human-Computer Interaction, UI, UX Design
- **Leadership: Director of Women in Science and Computing** (July 2022 - April 2023)

Projects

AI Chatbot for Personal Portfolio Website | *React, Tailwind CSS, React Router, GitHub Pages, Vercel, OpenAI API*

- Designed and developed a responsive pixel-art virtual desk UI portfolio with a production **RAG-based AI chatbot** backend using Node.js and OpenAI with semantic search, embeddings, caching, and serverless deployment on Vercel.

k-Nearest Neighbors (k-NN) MNIST Classifier | *Python, NumPy, Matplotlib, Pillow*

- Implemented a supervised learning algorithm to classify digits on a subset of 5,000 MNIST images, handling data loading, vectorization, distance computation, and normalization, achieving **90% test accuracy** on the best model.
- Visualized accuracy trends with Matplotlib across varying k values and normalization techniques, identifying optimal hyperparameters and improving model performance by 63% over the unnormalized baseline.

Three Musketeers | *Java, JavaFX, JUnit*

- Collaborated with a group of 3 students to develop a GUI board game using the MVC software design pattern and core OOP principles: abstraction, encapsulation, inheritance, polymorphism and unit testing with JUnit.

Heart Disease Prediction using Decision Trees (NHANES Dataset) | *Python, scikit-learn, Pandas, Matplotlib*

- Built and tuned Decision Tree models to predict heart disease risk on a NHANES dataset of 8,000 patient records, balancing bias–variance trade-offs and achieving **96.8%+ test accuracy** with optimal hyperparameters.
- Conducted EDA and feature engineering by encoding categorical variables, boosting validation accuracy by 25%.
- Visualized feature importance and decision boundaries, enhancing explainability for non-technical stakeholders.

Technical Skills

Languages: Python, Java, HTML, CSS, JavaScript, TypeScript, C, PostgreSQL, MS SQL, R, Bash, Shell
Frameworks: RESTful APIs, Spring Boot, React, Angular, NumPy, Pandas, Matplotlib, Scikit-learn, Pytest, JUnit
Dev Tools: Git, GitHub, VS Code, PyCharm, Eclipse, IntelliJ, SSMS, Jira, ALM, Linux, Figma, Adobe XD